A MODULE WITH A COMPLETE DEVISSAGE THAT IS FLAT AT THE POINT IS FINITELY PRESENTED, FREE OVER THE BASE AND A DIRECT SUM OF SMOOTH PIECES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Finiteness of presentation along a complete devissage, and localisation at the unique prime lying over the point. Same chapter flat.tex, label remark-finite-presentation (proof line 2447) and Algebra, label algebra-lemma-unique-prime-over-localize-below (proof line 2452). The first identifies finiteness of presentation of N over S with that of M_1 over B_1, and the second identifies the localisations at the two primes, which together let (S, N) be replaced by (B_1, M_1).
- The algebraic induction step for a flat map with geometrically integral fibres. Same chapter flat.tex, label lemma-induction-step, invoked at proof line 2456. Gives that the first map of the devissage is universally injective over the base and that the cokernel is flat at the point, which makes the induction hypothesis applicable to the cokernel.
- Finiteness of presentation of an extension, freeness of a projective module over a local ring, and projectivity of the smooth pieces. Algebra, labels algebra-lemma-extension (proof line 2465) and algebra-theorem-projective-free-over-local-ring (proof line 2471); Same chapter flat.tex, label lemma-fibres-irreducible-flat-projective-nonnoetherian (proof line 2470). Shows that M_1 is finitely presented as a B_1-module once its cokernel is, that the smooth algebra B_1 is free over the local base because it is projective, and that the decomposition of the cokernel supplies the required decomposition of M_1.

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Source reference. Original lemma 0519: sources/stacks-chow/tags.txt, source lines 6527–6527. Navigation pointer for \ref{algebra-lemma-extension} used inside the counted span (line 2465); the argument that consumes it is retained in the span.

Source reference. Original lemma 00EA: sources/stacks-chow/tags.txt, source lines 527–527. Navigation pointer for \ref{algebra-lemma-unique-prime-over-localize-below} used inside the counted span (line 2452); the argument that consumes it is retained in the span.

Source reference. Original theorem 0593: sources/stacks-chow/tags.txt, source lines 6801–6801. Navigation pointer for \ref{algebra-theorem-projective-free-over-local-ring} used inside the counted span (line 2472); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FT: sources/stacks-flat/pinned-original-source.tex, source lines 2247–2247. Navigation pointer for \ref{lemma-fibres-irreducible-flat-projective-nonnoetherian} used inside the counted span (line 2470); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I3: sources/stacks-flat/pinned-original-source.tex, source lines 2356–2356. Navigation pointer for \ref{lemma-induction-step} used inside the counted span (line 2456); the argument that consumes it is retained in the span.

Source reference. Original remark 05HY: sources/stacks-flat/pinned-original-source.tex, source lines 1685–1685. Navigation pointer for \ref{remark-finite-presentation} used inside the counted span (line 2447); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $(R, \mathfrak{m})$ be a local ring. Let $R \to S$ be a ring map of finite presentation. Let N be a finite S-module. Let $\mathfrak{q}$ be a prime of S lying over $\mathfrak{m}$. Assume that $N_\mathfrak{q}$ is flat over R, and assume there exists a complete dévissage of $N/S/R$ at $\mathfrak{q}$. Then N is a finitely presented S-module, free as an R-module, and there exists an isomorphism*

$$N \cong B_1^{\oplus r_1} \oplus \ldots \oplus B_n^{\oplus r_n}$$

as R-modules where each B_i is a smooth R-algebra with geometrically irreducible fibres.

Proof. Let $(A_i, B_i, M_i, \alpha_i, \mathfrak{q}_i)_{i=1,\ldots,n}$ be the given complete dévissage. We prove the lemma by induction on n. Note that N is finitely presented as an S-module if and only if M_1 is finitely presented as an B_1-module, see Remark 05HY. Note that $N_\mathfrak{q} \cong (M_1)_{\mathfrak{q}_1}$ as R-modules because (a) $N_\mathfrak{q} \cong (M_1)_{\mathfrak{q}_1'}$ where $\mathfrak{q}_1'$ is the unique prime in A_1 lying over $\mathfrak{q}_1$ and (b) $(A_1)_{\mathfrak{q}_1'} = (A_1)_{\mathfrak{q}_1}$ by Algebra, Lemma 00EA, so (c) $(M_1)_{\mathfrak{q}_1'} \cong (M_1)_{\mathfrak{q}_1}$. Hence $(M_1)_{\mathfrak{q}_1}$ is a flat R-module. Thus we may replace (S, N) by (B_1, M_1) in order to prove the lemma. By Lemma 05I3 the map $\alpha_1 : B_1^{\oplus r_1} \to M_1$ is R-universally injective and $\mathrm{Coker}(\alpha_1)_\mathfrak{q}$ is R-flat. Note that $(A_i, B_i, M_i, \alpha_i, \mathfrak{q}_i)_{i=2,\ldots,n}$ is a complete dévissage of $\mathrm{Coker}(\alpha_1)/B_1/R$ at $\mathfrak{q}_1$. Hence the induction hypothesis implies that $\mathrm{Coker}(\alpha_1)$ is finitely presented as a B_1-module, free as an R-module, and has a decomposition as in the lemma. This implies that M_1 is finitely presented as a B_1-module, see Algebra, Lemma 0519. It further implies that $M_1 \cong B_1^{\oplus r_1} \oplus \mathrm{Coker}(\alpha_1)$ as R-modules, hence a decomposition as in the lemma. Finally, B_1 is projective as an R-module by Lemma 05FT hence free as an R-module by Algebra, Theorem 0593. This finishes the proof. □